(a) What is the ground-state configuration of the valence electrons of bromine molecules (Br_2)? Is bromine paramagnetic or diamagnetic?

(b) What is the electron configuration of the Br_2^+ molecular ion? Is its bond stronger or weaker than that in Br_2? What is its bond order?

(c) Bromine compounds have been known and used for centuries. The deep purple color that symbolized imperial power in ancient Rome originated with the compound dibromoindigo, which was extracted in tiny quantities from purple snails (about 8000 snails per gram of compound). What color and maximum wavelength of *absorbed* light would give a deep purple (violet) color?

(d) What excited electronic state is responsible for the brownish-red color of bromine? Refer to Figure 6.16.

(e) The two naturally occurring isotopes of bromine are ^{79}Br and ^{81}Br, with masses of 78.918 and 80.916 u, respectively. The wavelength of the $J = 0$ to $J = 1$ rotational transition in $^{79}Br^{81}Br$ is measured to be 6.18 cm. Use this information to calculate the bond length in the Br_2 molecule, and compare the result with that listed in Table 3.3.

(f) The wavelength of the vibrational transition in the $^{79}Br^{81}Br$ molecule is 3.09×10^{-5} m. Calculate the force constant for the bond in this molecule.

(g) The action of light on bromine compounds released into the air (such as by leaded gasoline) causes the formation of the BrO radical. Give the bond order of this species by comparing it with the related radical OF.

(h) There is concern that synthetic bromine-containing compounds, in addition to chlorofluorocarbons, are helping to destroy ozone in the stratosphere. The BrO [see part (g)] can take part with ClO in the following catalytic cycle:

$$Cl + O_3 \longrightarrow ClO + O_2$$

$$Br + O_3 \longrightarrow BrO + O_2$$

$$ClO + BrO \longrightarrow Cl + Br + O_2$$

Write the overall equation for this cycle.

Answers

(a) $(\sigma_{g4s})^2(\sigma^*_{u4s})^2(\sigma_{g4p_z})^2(\pi_{u4p})^4(\pi^*_{g4p})^4$; diamagnetic

(b) $(\sigma_{g4s})^2(\sigma^*_{u4s})^2(\sigma_{g4p_z})^2(\pi_{u4p})^4(\pi^*_{g4p})^3$; stronger; bond order is $\frac{3}{2}$ versus 1

(c) Yellow light, near 530 nm (see Figure 4.3)

(d) The lowest energy excited state, which arises from the excitation of an electron from the filled π^*_{g4p} orbital to the unfilled σ^*_{u4pz} orbital

(e) 2.28 Å (from Table 3.3: 2.286 Å)

(f) 247 J Nm^{-1}

(g) $\frac{3}{2}$ order

(h) Overall: $2\ O_3 \longrightarrow 3\ O_2$

PROBLEMS

Answers to problems whose numbers are boldface appear in Appendix G. Problems that are more challenging are indicated with asterisks.

1. Stratospheric ozone absorbs ultraviolet radiation (UVB) that is known to cause cancer in humans. The onset of the UVB region of the spectrum is 320 nm and the absorption cross section for ozone (O_3) that wavelength is about 5×10^{-20} cm^2 $molecule^{-1}$ (see Fig. 20.44). The average ozone number density in the stratosphere is about 5×10^{12} molecules cm^{-3}. What thickness of the ozone layer is required to absorb 90% of the UV radiation at 320 nm?

2. The absorption cross section for fluorescein, a dye that is used as a fluorescent probe for biological imaging (see the Chapter Opener figure on page 941), is 9.25×10^{-16} cm^2 $molecule^{-1}$ at 500 nm. How much light is transmitted through a 1-cm path length by a standard 10^{-6} M solution of fluorescein used to label biological samples?

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3. The vibrational frequency ν for H_2 is 4156 cm^{-1} and its rotational constant $\tilde{B}$ is 0.082 cm^{-1}. Calculate the relative populations of the $v = 1$ and $v = 0$ vibrational energy levels and the relative populations of the $J = 1$ and $J = 0$ rotational levels for H_2 at the operating temperature of the space shuttle's combustion chamber, which is 3300 °C. Comment on your results.

4. The vibrational frequency ν for Br_2 is 323 cm^{-1} and its rotational constant $\tilde{B}$ is 60.96 cm^{-1}. Calculate the relative populations of the $v = 1$ and $v = 0$ vibrational energy levels and the relative populations of the $J = 1$ and $J = 0$ rotational levels for Br_2 at 300 K. Comment on your results.

Rotational and Vibrational Spectroscopy

5. Use data from Tables 19.1 and 3.3 to predict the energy spacing between the ground state and the first excited rotational state of the $^{14}N^{16}O$ molecule.

6. Use data from Tables 19.1 and 3.3 to predict the energy spacing between the ground state and the first excited rotational state of the $^{1}H^{19}F$ molecule.

7. The first three absorption lines in the pure rotational spectrum of gaseous $^{12}C^{16}O$ are found to have the frequencies 1.15×10^{11}, 2.30×10^{11}, and 3.46×10^{11} s^{-1}. Calculate:
   (a) The moment of inertia I of CO (in kg m^2)
   (b) The energies of the $J = 1$, $J = 2$, and $J = 3$ rotational levels of CO, measured from the $J = 0$ state (in joules)
   (c) The C—O bond length (in angstroms)

8. Four consecutive absorption lines in the pure rotational spectrum of gaseous $^{1}H^{35}Cl$ are found to have the frequencies 2.50×10^{12}, 3.12×10^{12}, 3.74×10^{12}, and 4.37×10^{12} s^{-1}. Calculate:
   (a) The moment of inertia I of HCl (in kg m^2)
   (b) The energies of the $J = 1$, $J = 2$, and $J = 3$ rotational levels of HCl, measured from the $J = 0$ state (in joules)
   (c) The H—Cl bond length (in angstroms)
   (d) The initial and final J states for the observed absorption lines

9. The rotational constant $\tilde{B} = 1.45$ cm^{-1} for O_2. Calculate the O_2 bond length and predict the appearance of its rotational Raman spectrum, recalling that transitions between states with even J are forbidden due to nuclear spin statistics considerations.

10. The rotational constant $\tilde{B}$ of $^{7}Li_2$ is 0.6727 cm^{-1}. Calculate the $^{7}Li_2$ bond length and predict the appearance of its rotational Raman spectrum without the effect of nuclear spin statistics on the relative intensities of the lines. Compare your results with those of Example 20.4.

11. In Example 20.3, we determined that the moment of inertia of the NaH molecule is 5.70×10^{-47} kg m^2.
   (a) Calculate the relative population of the $J = 5$ level and the ground state at 25°C.
   (b) Calculate the relative population of the $J = 15$ level and the ground state at 25°C.
   (c) Calculate the relative population of the $J = 25$ level and the ground state at 25°C.

12. The rotational constant for $^{14}N_2$, measured for the first time using rotational Raman spectroscopy (see Fig. 20.8), is 1.99 cm^{-1}. Calculate (a) the moment of inertia I of N_2 in kg m^2 (b) The energies of the first three excited rotational levels, relative to that of the ground level, in wave numbers (c) The frequency of the $J = 0 \longrightarrow J = 1$ transition in GHz.

13. The Li_2 molecule (^{7}Li isotope) shows a very weak infrared line in its vibrational spectrum at a wavelength of 2.85×10^{-5} m. Calculate the force constant for the Li_2 molecule.

14. The Na_2 molecule (^{23}Na isotope) shows a very weak infrared line in its vibrational spectrum at a wavelength of 6.28×10^{-5} m. Calculate the force constant for the Na_2 molecule, and compare your result with that of problem 13. Give a reason for any difference.

15. The "signature" infrared absorption that indicates the presence of a C—H stretching motion in a molecule occurs at wavelengths near 3.4×10^{-6} m. Use this information to estimate the force constant of the C—H stretch. Take the reduced mass in this motion to be approximately equal to the mass of the hydrogen atom (a good approximation when the H atom is attached to a heavy group).

16. Repeat the calculation of the preceding problem for the N—H stretch, where absorption occurs near 2.9×10^{-6} m. Which bond is stiffer: N—H or C—H?

17. Estimate the ratio of the number of molecules in the first excited vibrational state of the molecule N_2 to the number in the ground state, at a temperature of 450 K. The vibrational frequency of N_2 is 7.07×10^{13} s^{-1}.

18. The vibrational frequency of the ICl molecule is 1.15×10^{13} s^{-1}. For every million (1.00×10^{6}) molecules in the ground vibrational state, how many will be in the first excited vibrational state at a temperature of 300 K?

19. The measured vibrational frequencies for the hydrogen halides given in Section 20.3 are: $^{1}H^{19}F$ (4139 cm^{-1}); $^{1}H^{35}Cl$ (2991 cm^{-1}); $^{1}H^{81}Br$ (2449 cm^{-1}); and $^{1}H^{127}I$ (2308 cm^{-1}). Calculate the force constants for each molecule, using the vibrational frequencies given and the atomic masses provided in Table 19.1. Which is the more important factor in determining the vibrational frequencies in this series of molecules, the force constants or the reduced masses?

20. The vibrational frequencies for the mixture of gases shown in the Chapter 3 *Connection to Instrumental Analysis* include: H_2 (4160 cm^{-1}) N_2 (2331 cm^{-1}) CO (2143 cm^{-1}), and O_2 (1550 cm^{-1}). What can you conclude about the bond orders by inspection? Calculate the force constants for each molecule to check the accuracy of your estimate.

21. Substitution of hydrogen by deuterium is a convenient way to label organic molecules at specific sites for the purpose of investigating reaction mechanisms. At what frequency would the C—D stretch of a phenyl ring appear if the frequency of the C—H stretch is 3062 cm^{-1}?

22. ^{13}C is also used as a label in organic chemistry. What is the frequency of a $^{12}C—^{13}C$ single bond stretch if the frequency of the corresponding $^{12}C—^{12}C$ stretch appears at 1053 cm^{-1}?

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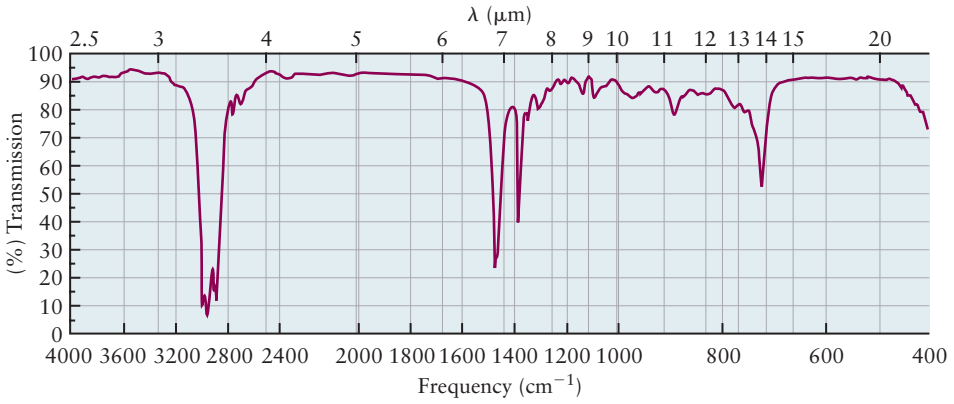
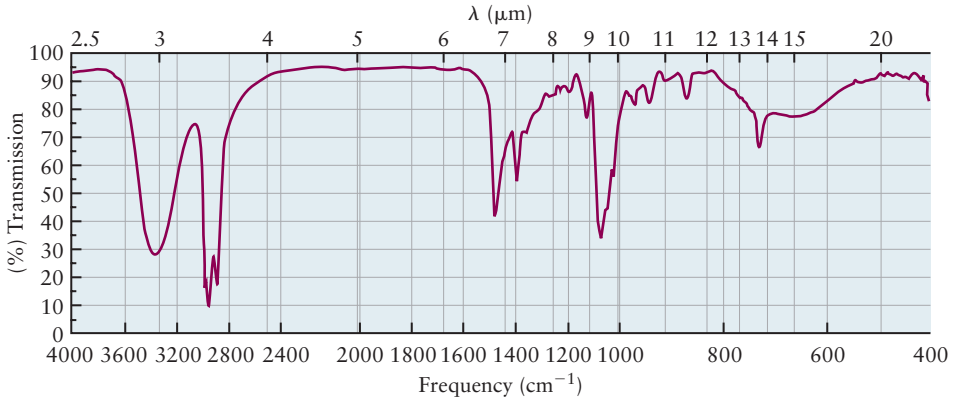
23. How many normal vibrational modes does each of the following molecules have?

(a) NH_3
(b) C_2H_4
(c) CCl_2F_2
(d) CH_3CH_2OH

24. Consider the vibrations of the molecules in the previous problem from a localized point of view, as the vibrations of individual bonds. Infrared absorption is proportional to the permanent dipole moment of each individual bond. Which of the molecules will absorb IR radiation most strongly and why?

25. The IR spectra of nonane (C_9H_{20}) and 1-hexanol ($C_6H_{13}OH$) are shown below. Assign each spectrum to the correct compound and identify the frequencies and the functional groups used to support your assignment.





26. The IR spectrum of 2-methyl-1-butanol ($H_3CCH_2CH_2(CH_3)CH_2OH$) and *tert*-butyl methyl ether [$(CH_3)_3OCH_3$] are shown on p. 1027. Assign each spectrum to the correct compound and identify the frequencies and the functional groups used to support your assignment.

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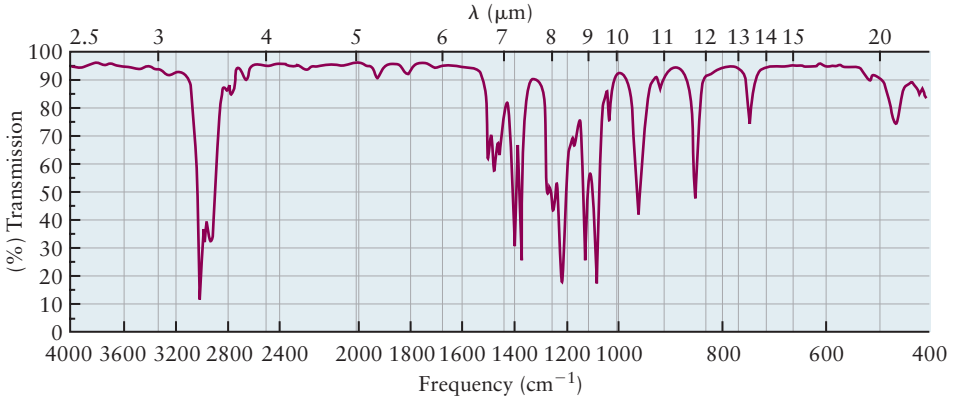

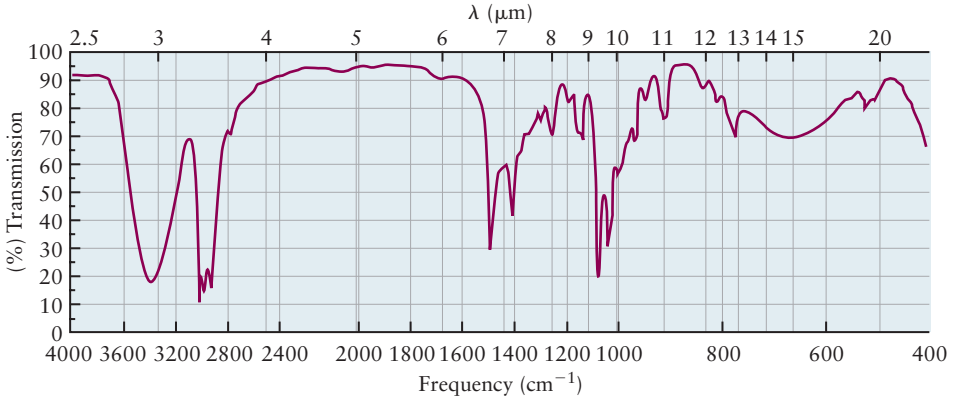

Nuclear Magnetic Resonance Spectroscopy

27. Give the number of peaks and the relative peak areas that should be observed in the low-resolution proton magnetic resonance spectra of the following molecules:

$$CH_3CH_2CH_2CH_3,\ CH_3OCH_3,\ CH_3NHCH_3.$$

28. The organic compound 1,4-dimethylbenzene (also known as *p*-xylene) has the formula $(CH_3)_2C_6H_4$. Its structure has two CH_3 (methyl) groups substituted at opposite positions on the benzene (C_6H_6) ring. Predict the number of peaks in the low-resolution proton NMR spectrum of this compound and the relative areas of the peaks.

29. The NMR spectrum of a molecule closely related to that discussed in Example 20.9 has the molecular formula $C_7H_{14}O$. The ^1H NMR spectrum shows three single peaks at $\delta = 1.01$, 2.11, and 2.32 with peak areas in the ratio 9:3:2, respectively. What is the structure of the molecule?

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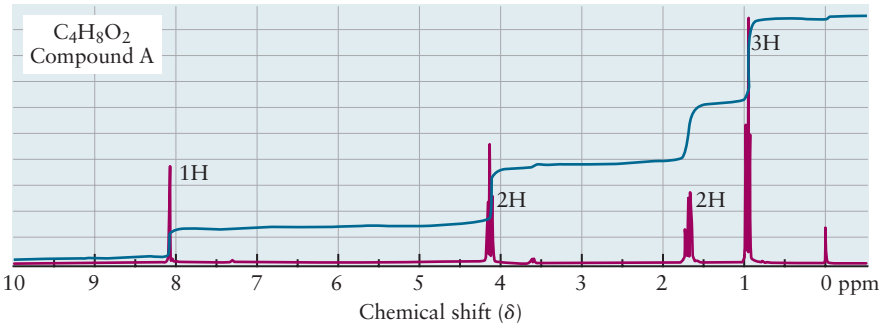
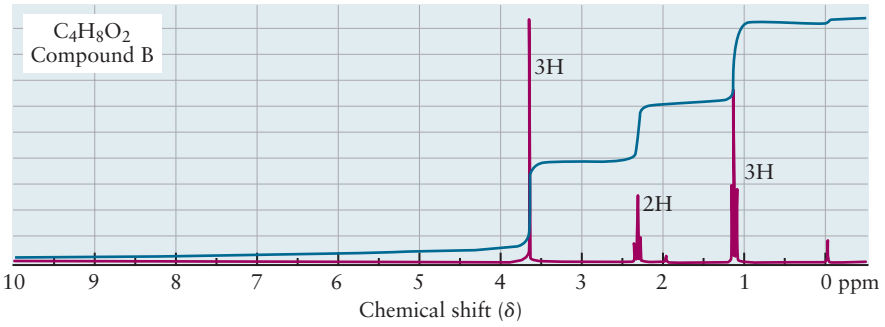
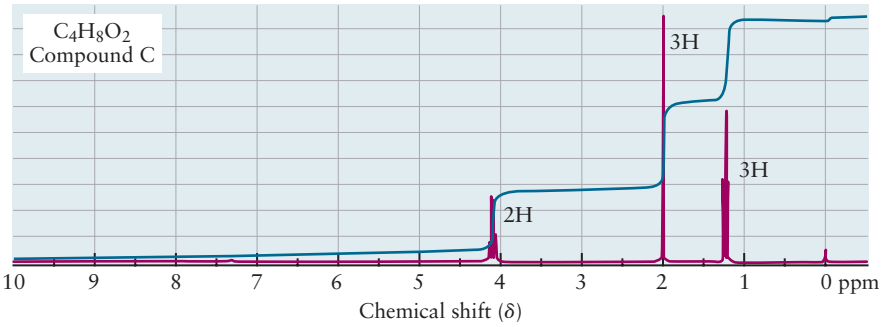
30. Identify the three molecules with the molecular formula $C_4H_8O_2$ from their NMR spectrum shown below. Hint: they are all esters.







Introduction to Molecular Spectroscopy

31. The percentage transmittance of light at 250 nm through a certain aqueous solution is 20.0% at 25°C. The experimental cell length is 1.0 cm, and the concentration of the solution is 5×10^{-4} mol L^{-1}. Calculate the absorbance. Calculate the molar absorption coefficient.

32. Beer's law is used to measure the concentration of species in solutions, once a "standardization curve" has been prepared for that species. In one such experiment, percent transmission was measured for a series of solutions with known concentrations and the results were as follows:

Concentration μg mL^{-1}	1.0	2.0	3.0	4.0	5.0
Transmission, percent	66.8	44.7	29.2	19.9	13.3

Plot these results to obtain the standardization curve. An unknown concentration of the same species, measured in the same transmission cell, transmitted 35% of the incoming light. Calculate the concentration of the unknown solution.

33. Beer's law can be used to determine the concentration of two substances *A* and *B* in solution, provided they do not react or interact, so they absorb radiation independently. The following data were obtained for *A* and *B* in three different solutions:

	[A] mol L^{-1}	[B] mol L^{-1}	% Transmittance at λ = 400 nm	% Transmittance at λ = 500 nm
Solution 1	0.0010	0	10.0	60.0
Solution 2	0	0.0050	80.0	20.0
Solution 3	?	?	40.0	50.0

Calculate the concentrations of A and B in Solution 3.

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34. The absorption of ultraviolet light by proteins at wavelength 280 nm is caused mostly by the amino acids tyrosine and tryptophan along the protein molecular chains. The molecular absorption coefficients for these two amino acids are:

$$\epsilon_{tryp}^{280} = 5690 \text{ L cm}^{-1} \text{ mol}^{-1}$$
$$_{tyro}^{280} = 1280 \text{ L cm}^{-1} \text{ mol}^{-1}$$

Experiments are carried out on a protein with molecular weight 26,000, which contains two units of tryptophan and six units of tyrosine along the chain. The absorption is measured in a cell 1 cm long, and the protein concentration is 1.0 mg mL^{-1}. Calculate the absorbance and the percent transmission.

Electronic Spectroscopy and Excited State Relaxation Processes

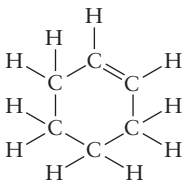
35. Suppose that an ethylene molecule gains an additional electron to give the $C_2H_4^-$ ion. Will the bond order of the carbon–carbon bond increase or decrease? Explain.

36. Suppose that an ethylene molecule is ionized by a photon to give the $C_2H_4^+$ ion. Will the bond order of the carbon–carbon bond increase or decrease? Explain.

37. The color of the dye "indanthrene brilliant orange" is evident from its name. In what wavelength range would you expect to find the maximum in the absorption spectrum of this molecule? Refer to the color spectrum in Figure 4.3.

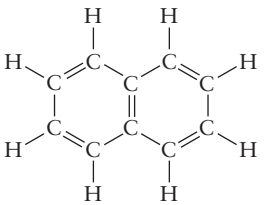
38. In what wavelength range would you expect to find the maximum in the absorption spectrum of the dye "crystal violet"?

39. The structure of the molecule cyclohexene is shown below:



Does the absorption of ultraviolet light by cyclohexene occur at shorter wavelengths than in benzene? Explain.

40. The naphthalene molecule has a structure that corresponds to two benzene molecules fused together:



The π-electrons in this molecule are delocalized over the entire molecule. The wavelength of maximum absorption in the UV-visible part of the spectrum in benzene is 255 nm. Is the corresponding wavelength shorter or longer than 255 nm for naphthalene?

41. Use data from Table 3.3 to give an upper bound on the wavelengths of light that are capable of dissociating a molecule of ClF.

42. Use data from Table 3.3 to give an upper bound on the wavelengths of light that are capable of dissociating a molecule of ICl.

Introduction to Atmospheric Chemistry

43. The bond dissociation energy of a typical C—F bond in a chlorofluorocarbon is approximately 440 kJ mol^{-1}. Calculate the maximum wavelength of light that can photodissociate a molecule of CCl_2F_2, breaking such a C—F bond.

44. The bond dissociation energy of a typical C—Cl bond in a chlorofluorocarbon is approximately 330 kJ mol^{-1}. Calculate the maximum wavelength of light that can photodissociate a molecule of CCl_2F_2, breaking such a C—Cl bond.

45. Draw a Lewis diagram(s) for the ozone molecule (O_3). Determine the steric number and hybridization of the central oxygen atom, and identify the molecular geometry. Describe the nature of the π bonds and give the bond order of the O—O bonds in ozone.

46. The compounds carbon dioxide (CO_2) and sulfur dioxide (SO_2) are formed by the burning of coal. Their apparently similar formulas mask underlying differences in molecular structure. Determine the shapes of these two types of molecules, identify the hybridization at the central atom of each, and compare the natures of their π bonds.

Photosynthesis

47. One way in which photosynthetic bacteria store chemical energy is through the conversion of a compound called adenosine diphosphate (ADP), together with hydrogen phosphate ion, to adenosine triphosphate (ATP):

$$ADP^{3-} + HPO_4^{2-} + H_3O^+ \longrightarrow ATP^{4-} + 2\ H_2O$$
$$\Delta G = +34.5 \text{ kJ (pH 7)}$$

Suppose some chlorophyll molecules absorb 1.00 mol of photons of blue light with wavelength 430 nm. If *all* this energy could be used to convert ADP to ATP at room conditions and pH 7, how many molecules of ATP would be produced per photon absorbed? (The actual number is smaller because the conversion is not 100% efficient.)

48. Repeat the calculation of the preceding problem for red light with wavelength 700 nm.

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ADDITIONAL PROBLEMS

49. What are the moments of inertia of $^{1}H^{19}F$ and $^{1}H^{81}Br$, expressed in kg m^2? Compute the spacings $\nu = \Delta E/h$ of the rotational states, in s^{-1}, between $J = 0$ and 1 and between $J = 1$ and 2. Explain, in one sentence, why the large change in mass from 19 to 81 causes only a small change in rotational energy differences.

50. The average bond length of a molecule can change slightly with vibrational state. In $^{23}Na^{35}Cl$, the frequency of light absorbed in a change from the $J = 1$ to the $J = 2$ rotational state in the ground vibrational state ($n = 0$) was measured to be $\nu = 2.60511 \times 10^{10}$ s^{-1}, and that for a change from $J = 1$ to $J = 2$ in the first excited vibrational state ($n = 1$) was $\nu = 2.58576 \times 10^{10}$ s^{-1}. Calculate the average bond lengths of NaCl in these two vibrational states, taking the relative atomic mass of ^{23}Na to be 22.9898 and that of ^{35}Cl to be 34.9689.

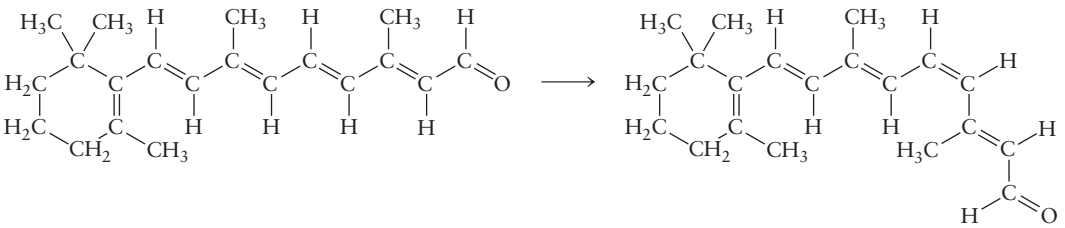
51. The vibrational frequencies of $^{23}Na^{1}H$, $^{23}Na^{35}Cl$, and $^{23}Na^{127}I$ are 3.51×10^{13} s^{-1}, 1.10×10^{13} s^{-1}, and 0.773×10^{13} s^{-1}, respectively. Their bond lengths are 1.89 Å, 2.36 Å, and 2.71 Å. What are their reduced masses? What are their force constants? If NaH and NaD have the same force constant, what is the vibrational frequency of NaD? D is ^{2}H.

52. Recall that nuclear spin states in nuclear magnetic resonance are typically separated by energies of 2×10^{-5} kJ mol^{-1} to 2×10^{-4} kJ mol^{-1}. What are the ratios of occupation probability between a pair of such levels at thermal equilibrium and a temperature of 25°C?

53. The vibrational temperature of a molecule prepared in a supersonic jet can be estimated from the observed populations of its vibrational levels, assuming a Boltzmann distribution. The vibrational frequency of HgBr is 5.58×10^{12} s^{-1}, and the ratio of the number of molecules in the $n = 1$ state to the number in the $n = 0$ state is 0.127. Estimate the vibrational temperature under these conditions.

54. An electron in the π orbital of ethylene (C_2H_4) is excited by a photon to the π^* orbital. Do you expect the equilibrium bond length in the excited ethylene molecule to be greater or less than that in ground-state ethylene? Will the vibrational frequency in the excited state be higher or lower than in the ground state? Explain your reasoning.

55. One isomer of retinal is converted to a second isomer by the absorption of a photon:



This process is a key step in the chemistry of vision. Although free retinal (in the form shown to the left of the arrow) has an absorption maximum at 376 nm, in the ultraviolet region of the spectrum this absorption shifts into the visible range when the retinal is bound in a protein, as it is in the eye.

(a) How many of the C=C bonds are *cis* and how many are *trans* in each of the preceding structures? (When assigning labels, consider the relative positions of the two largest groups attached at each double bond.) Describe the motion that takes place upon absorption of a photon.

(b) If the ring and the —CHO group in retinal were replaced by $-CH_3$ groups, would the absorption maximum in the molecule shift to longer or shorter wavelengths?

* 56. The ground-state electron configuration of the H_2^+ molecular ion is $(\sigma_{g1s})^1$.

(a) A molecule of H_2^+ absorbs a photon and is excited to the σ^*_{u1s} molecular orbital. Predict what happens to the molecule.

(b) Another molecule of H_2^+ absorbs even more energy in an interaction with a photon and is excited to the σ_{g2s} molecular orbital. Predict what happens to this molecule.

* 57. (a) Draw a Lewis diagram for formaldehyde (H_2CO), and decide the hybridization of the central carbon atom.

(b) Formulate the molecular orbitals for the molecule.

(c) A strong absorption is observed in the ultraviolet region of the spectrum and is attributed to a $\pi \longrightarrow \pi^*$ transition. Another, weaker transition is observed at lower frequencies. What electronic excitation causes the weaker transition?

58. Write balanced chemical equations that describe the formation of nitric acid and sulfuric acid in rain, starting with the sulfur in coal and the oxygen, nitrogen, and water vapor in the atmosphere.

59. Compare and contrast the roles of ozone (O_3) and nitrogen dioxide (NO_2) in the stratosphere and in the troposphere.

60. Describe the greenhouse effect and its mechanism of operation. Give three examples of energy sources that contribute to increased CO_2 in the atmosphere and three that do not.

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61. Draw a schematic diagram of the steps in bacterial photosynthesis, numbering them in sequence and showing the approximate spatial relations of the involved molecules.

62. Do you expect the energy of the special pair of bacteriochlorophyll molecules to be higher or lower than the energy of an isolated bacteriochlorophyll? (*Hint:* Think about the analogy between the mixing of atomic orbitals to make molecular orbitals and the mixing of molecular orbitals on two nearby molecules.)

CUMULATIVE PROBLEMS

63. At thermal equilibrium, is the rate as a molecule is excited from $n = 0$ to the $n = 1$ level greater than or less than the rate for the reverse process? What is the ratio of the rate constants? (*Hint:* Think of the analogy with the chemical equilibrium between two species.)

* 64. It is important to know the dissociation constant of an indicator in order to use it properly in acid–base titrations. Spectrophotometry can be used to measure the concentration of these intensely colored species in acidic versus basic solutions, and from these data the equilibrium between the acidic and basic forms can be calculated. In one such study on the indicator *m*-nitrophenol, a 6.36×10^{-4} M solution was examined by spectrophotometry at 390 nm and 25°C in the following experiments. In highly acidic solution, where essentially all the indicator was in the form HIn, the absorbance was 0.142. In highly basic solution, where essentially all of the indicator was in the form In^-, the absorbance was 0.943. In a further series of experiments, the pH was adjusted using a buffer solution of ionic strength I, and absorbance was measured at each pH value. The following results were obtained:

pH	I	A
8.321	0.10	0.527
8.302	0.08	0.518
8.280	0.06	0.505
8.251	0.04	0.493
8.207	0.02	0.470

Calculate pK_a for the indicator at each value of ionic strength.

65. The hydroxyl radical has been referred to as the "chief clean-up agent in the troposphere." Its concentration is approximately zero at night and becomes as high as 1×10^7 molecules per cm^3 in highly polluted air.
   (a) Calculate the maximum mole fraction and partial pressure of OH in polluted air at 25°C and atmospheric pressure.
   (b) Write an equation for the reaction of HO with NO_2 in the atmosphere. How does the oxidation state of nitrogen change in this reaction? What is the ultimate fate of the product of this reaction?

66. In unpolluted air at 300 K, the hydroxyl radical OH reacts with CO with a bimolecular rate constant of 1.6×10^{11} L mol^{-1} s^{-1} and with CH_4 with a rate constant of 3.8×10^9 L mol^{-1} s^{-1}. Take the partial pressure of CO in air to be constant at 1.0×10^{-7} atm and that of CH_4 to be 1.7×10^{-6} atm, and assume that these are the primary mechanisms by which OH is consumed in the atmosphere. Calculate the half-life of OH under these conditions.

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